

FMCG SUPPLY CHAIN WAREHOUSE

PERFORMANCE ANALYSIS



BUSINESS PROBLEMS



Uneven shipment distribution across warehouses

Limited visibility into warehouse performance

Difficult to optimize inventory allocation

Need data-driven insights for operational improvement

BUSINESS QUESTIONS



Which regions and zones handle the highest and lowest shipment volumes ?

How do operational issues impact shipment volume?

Does warehouse capacity affect the amount of products shipped?

Do warehouse infrastructure and facility conditions affect shipment performance?

Do the numbers of retail shops and distributors influence shipment volume?

Does the distance from the warehouse to the distribution hub influence shipment volume?

Is the number of refill requests in the last three months related to shipment volume?

What characteristics are commonly found in warehouses with the highest shipment volumes?

DATASET & ANALYSIS WORKFLOW

Dataset Overview

 SURAJ9727 · UPDATED 3 YEARS AGO

12 <> Code Download

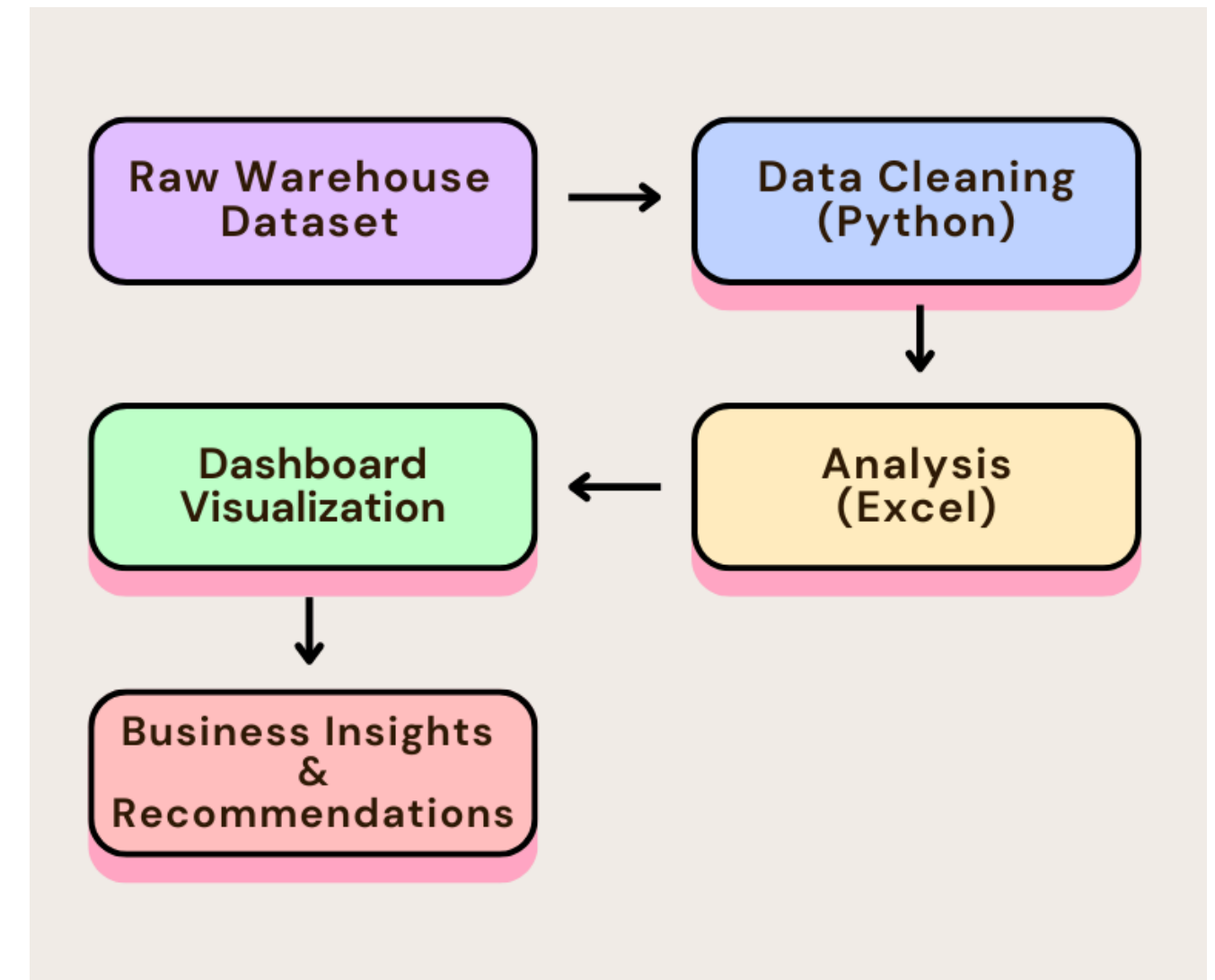
Supply Chain Optimization for a FMCG Company

Data Analysis/ Model Building/ Supply Chain Optimization



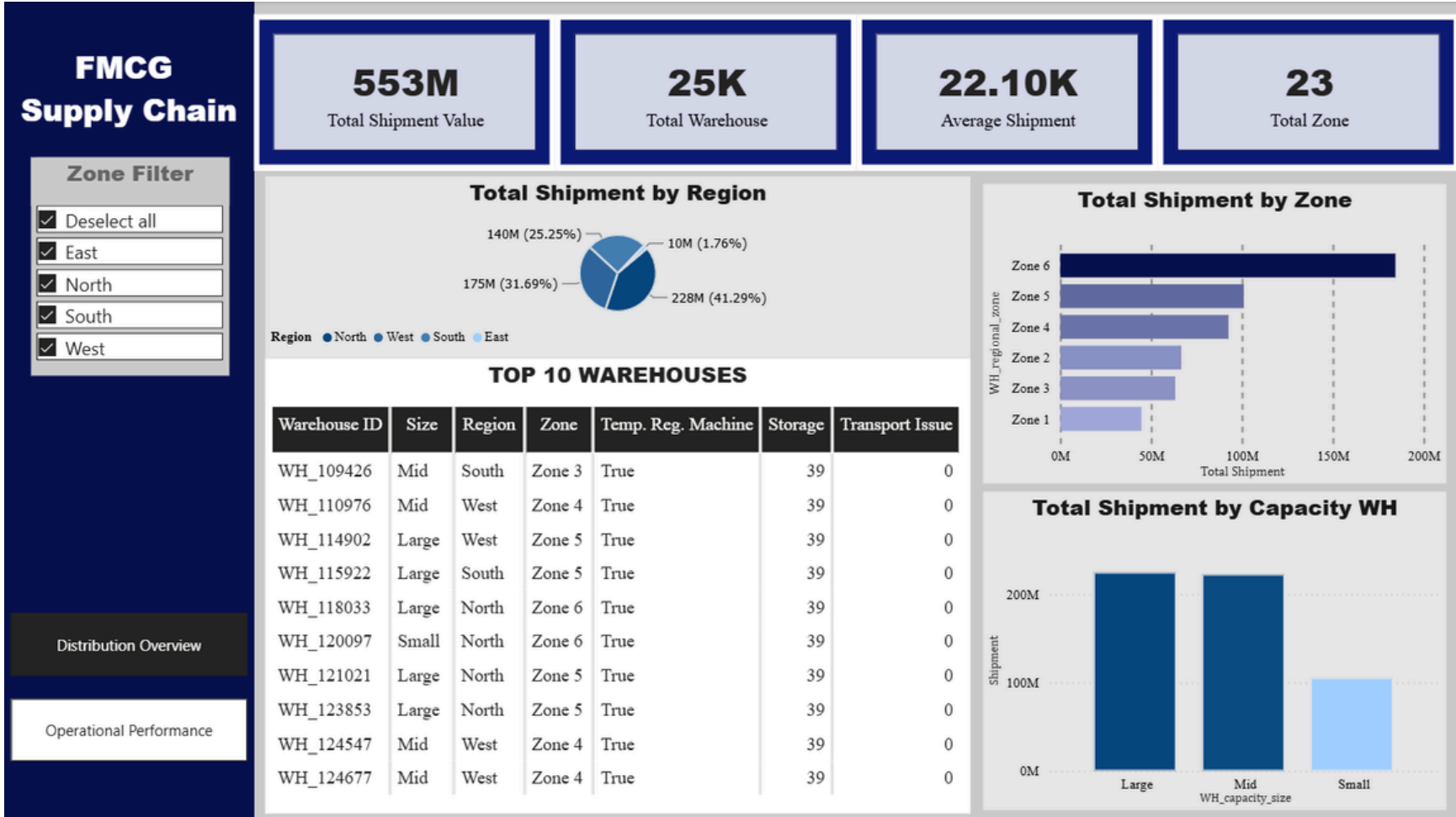
Dataset FMCG Supply Chain Warehouse
25,000 Warehouse Record

Analysis Workflow

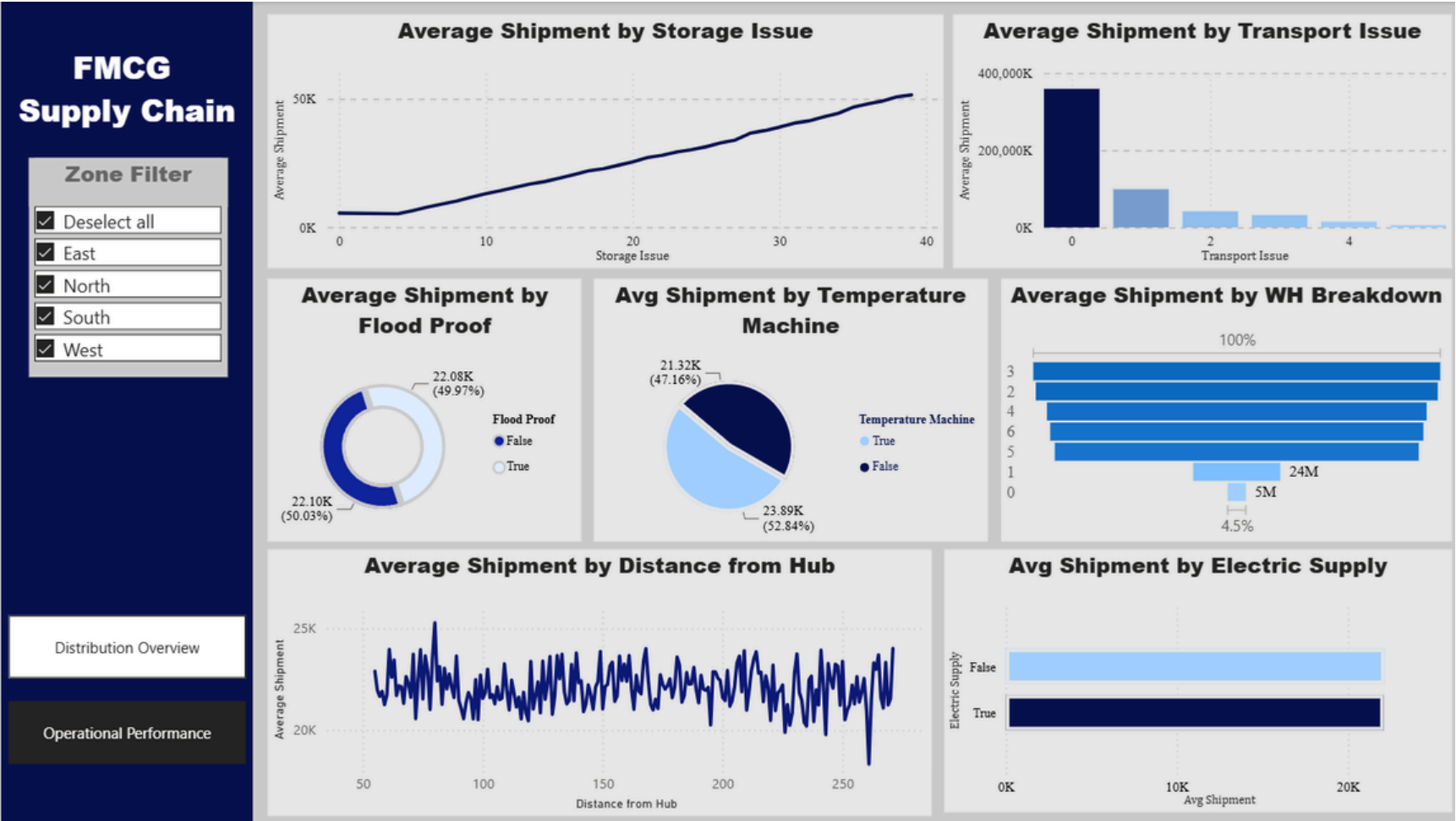


DASHBOARD OVERVIEW

Distribution Overview

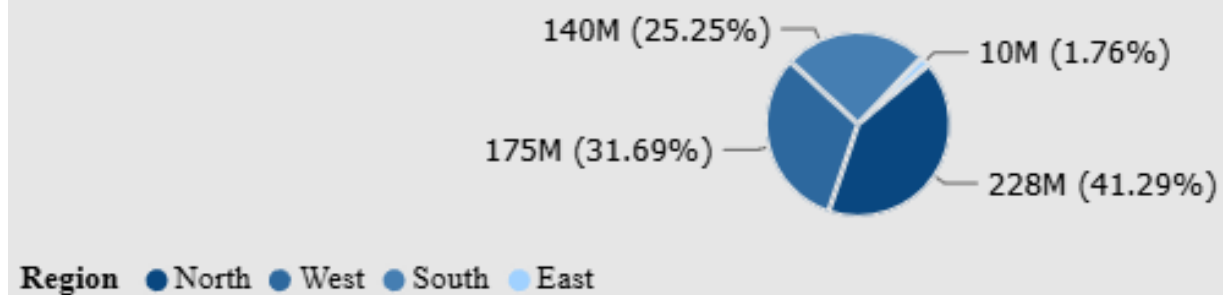


Operational Performance

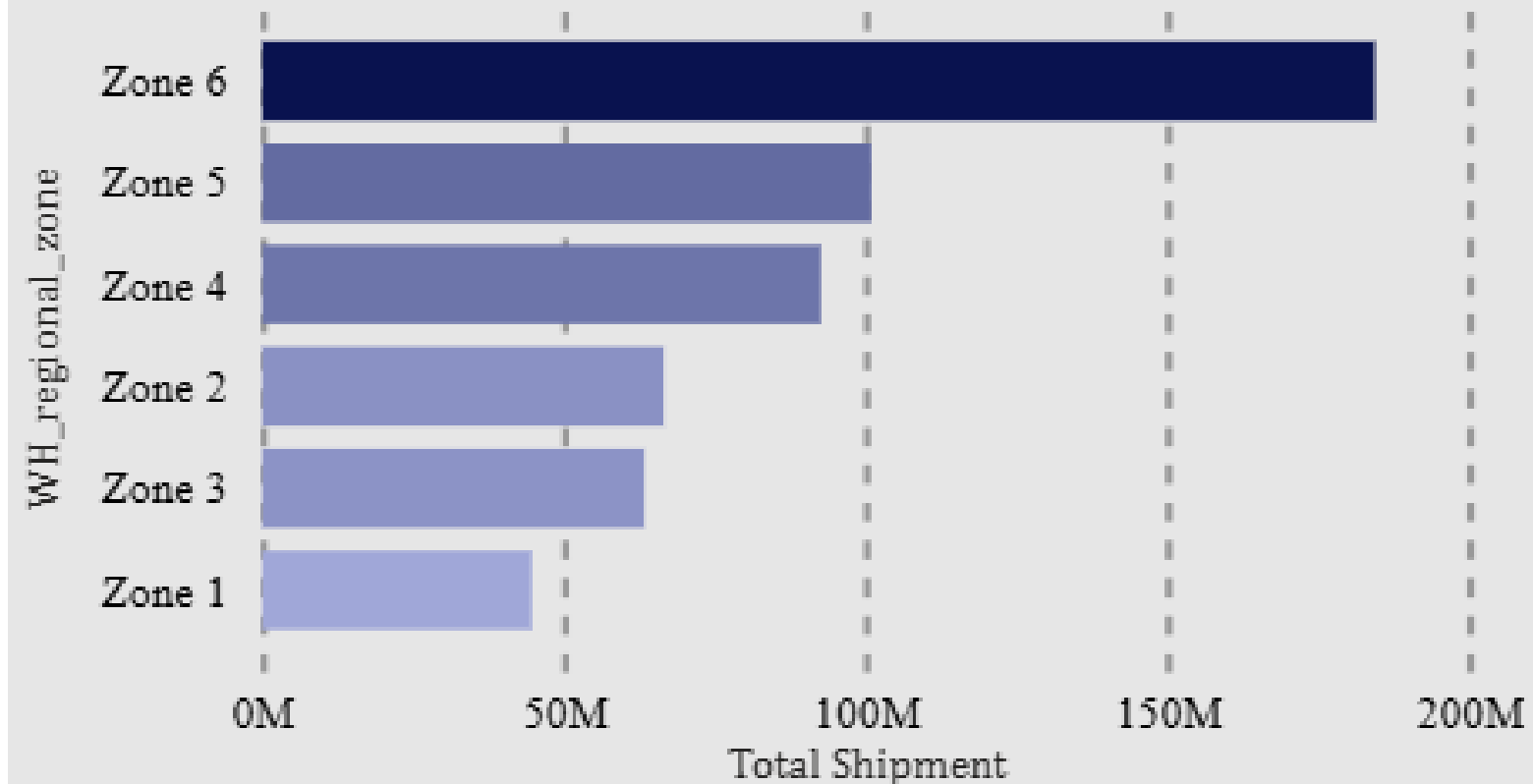


REGIONAL PERFORMANCE

Total Shipment by Region



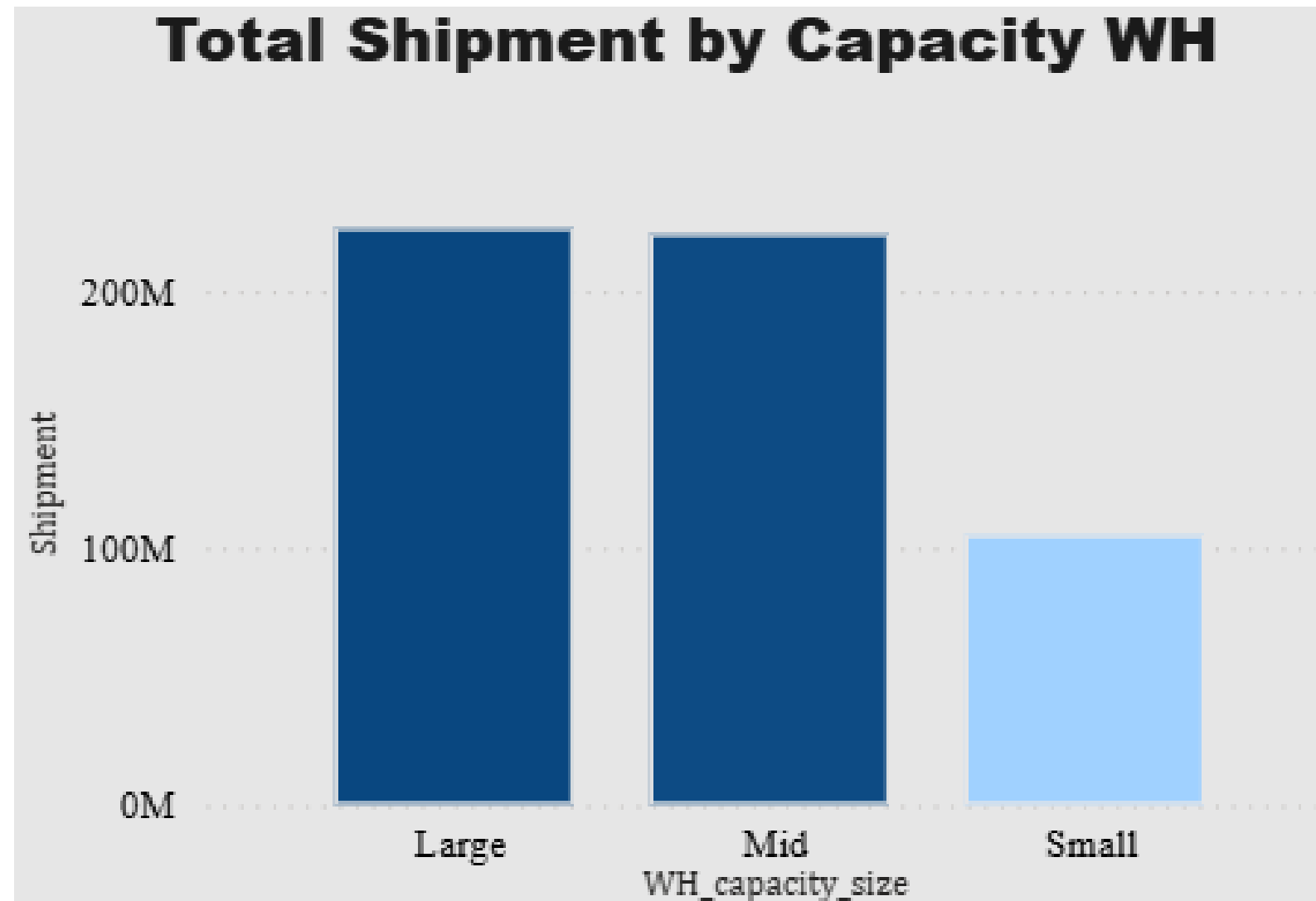
Total Shipment by Zone



Key Insights

- North contributed the highest shipment volume
- East recorded the lowest shipment volume
- Shipment distribution is highly imbalanced
- Zone 6 dominated shipments in North and West

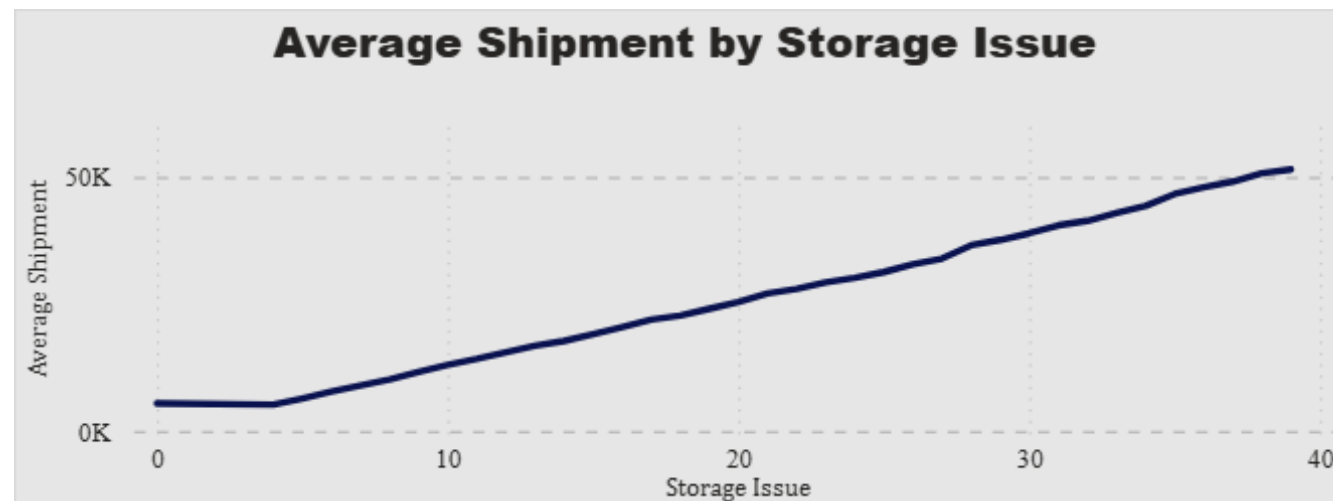
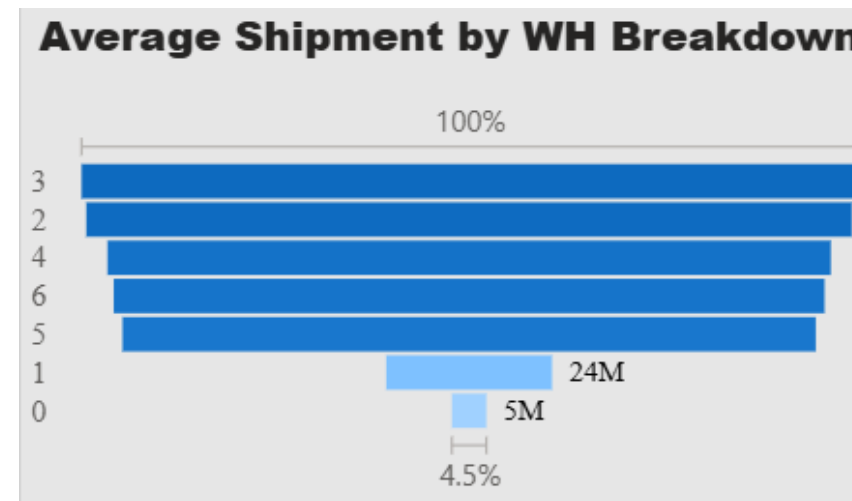
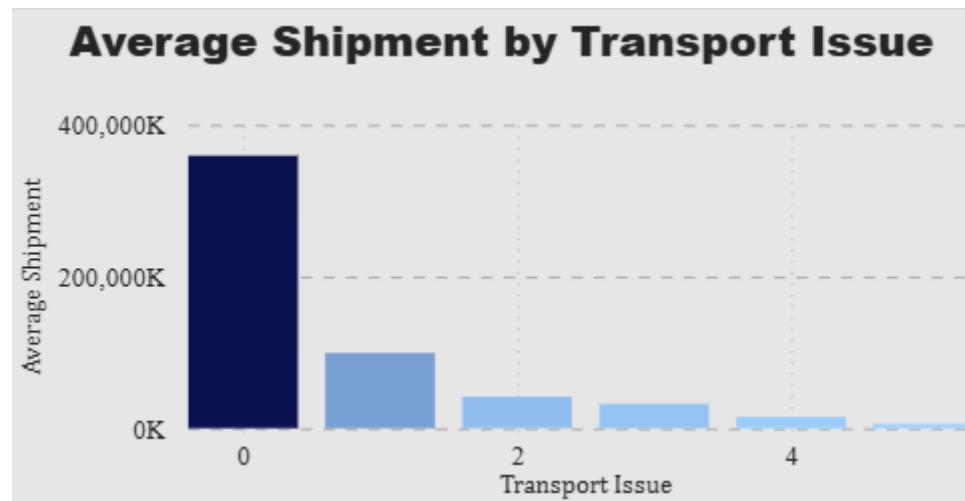
WAREHOUSE CAPACITY ANALYSIS



Key Insights

- Large warehouses generated the highest total shipment
- Average shipment differences were minimal
- Capacity alone does not determine shipment performance

OPERATIONAL FACTORS

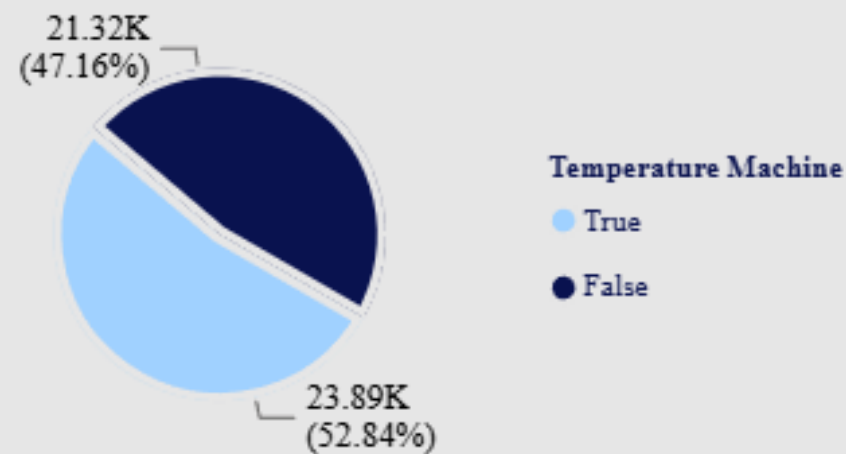


Key Insights

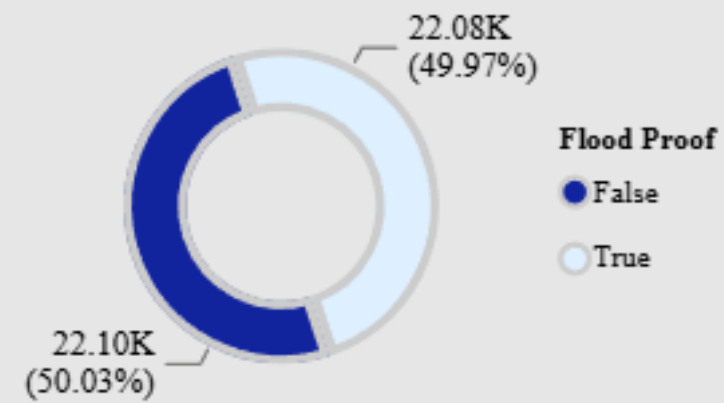
- High-volume warehouses reported more storage issues
- Breakdown frequency increased with warehouse utilization
- Transportation issues slightly reduced shipment volume

INFRASTRUCTURE ANALYSIS

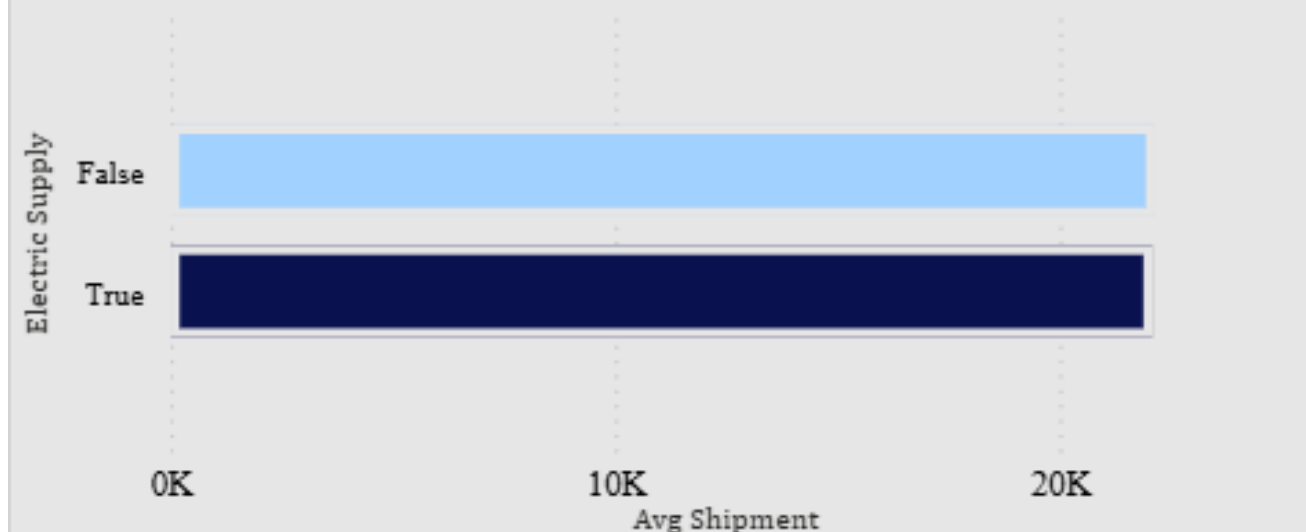
Avg Shipment by Temperature Machine



Average Shipment by Flood Proof



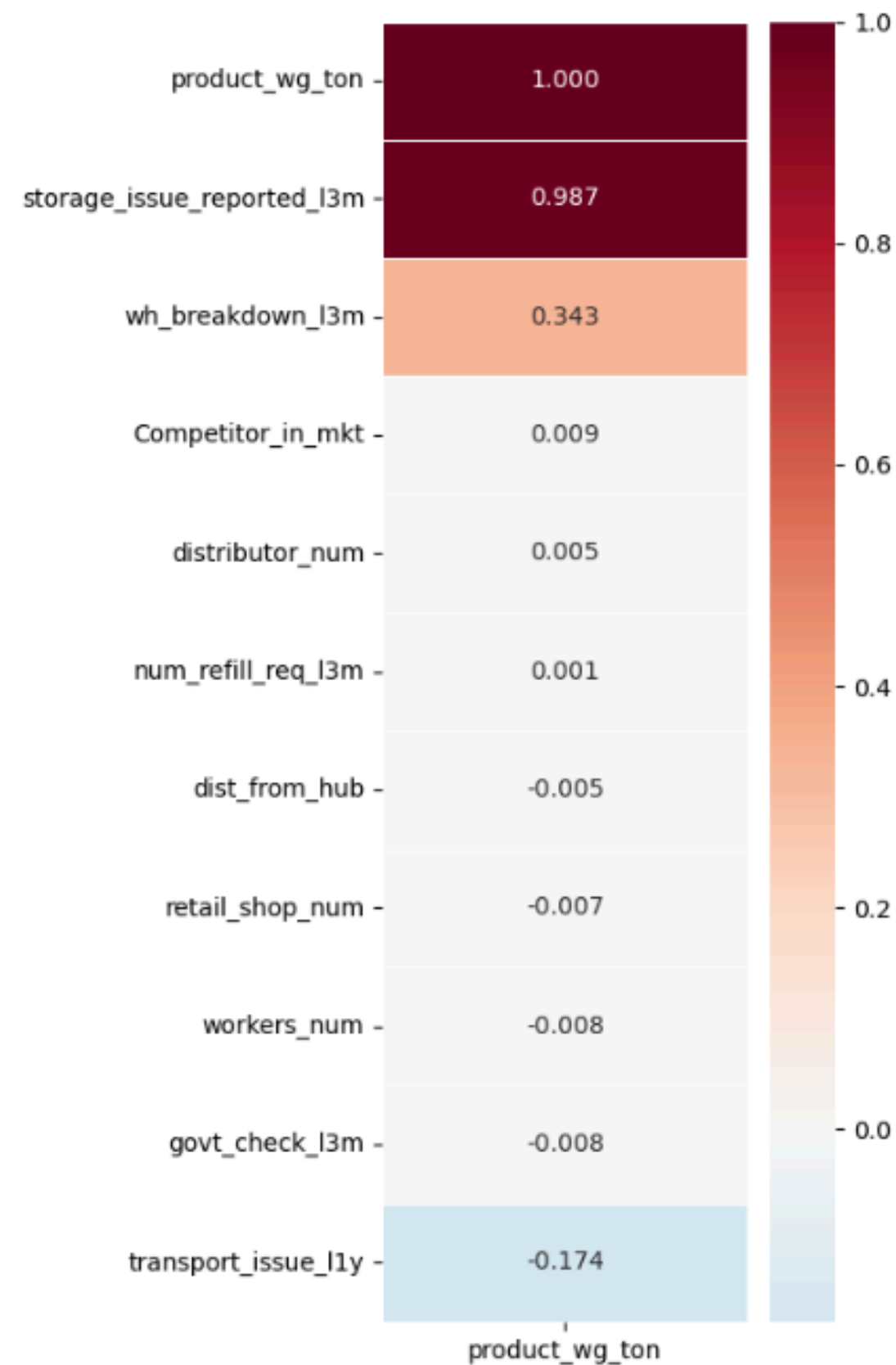
Avg Shipment by Electric Supply



Key Insights

- Temperature-controlled warehouses shipped approximately 12% more products
- Electric supply showed minimal impact
- Flood-proof facilities had little influence

CORRELATION ANALYSIS



Key Insights

- Refill requests shops showed almost no correlation.
- Distributor numbers had negligible influence.
- Retail were weakly related.
- Distance from hub was not a significant factor.

TOP PERFORMING WAREHOUSES

TOP 10 WAREHOUSES

Warehouse ID	Size	Region	Zone	Temp. Reg. Machine	Storage	Transport Issue
WH_109426	Mid	South	Zone 3	True	39	0
WH_110976	Mid	West	Zone 4	True	39	0
WH_114902	Large	West	Zone 5	True	39	0
WH_115922	Large	South	Zone 5	True	39	0
WH_118033	Large	North	Zone 6	True	39	0
WH_120097	Small	North	Zone 6	True	39	0
WH_121021	Large	North	Zone 5	True	39	0
WH_123853	Large	North	Zone 5	True	39	0
WH_124547	Mid	West	Zone 4	True	39	0
WH_124677	Mid	West	Zone 4	True	39	0

Characteristics

- Have Temperature regulation machine
- Mid / Large capacity
- North & West regions
- No transportation issues
- High storage activity

BUSINESS STRATEGY

Balance Warehouse Utilization

Reallocate inventory and workloads to improve utilization across warehouses, especially in underutilized regions.

Reduce Transportation Issues

Improve transportation reliability to minimize shipment delays and support higher shipment performance.

Evaluate Temperature-Controlled Facilities

Assess whether expanding temperature controlled facilities can improve shipment performance in strategic warehouses.

Adopt Best Practices

Apply successful operational practices from high-performing warehouses to lower performing ones.

Focus on Operational Efficiency

Prioritize operational improvements over warehouse expansion, as warehouse size alone has little impact on shipment performance.

Investigate the East Region

Further analyze the East region to identify the reasons behind its consistently low shipment volume.

IMPORTANCE OF **SUPPLY CHAIN**

Reduce operational costs

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Speed up delivery

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Improve customer satisfaction

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Reduce the risk of delays

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THANK YOU !

TOOLS USED : EXCEL | PYTHON | POWER BI

